KALINGA AI: A NATIONAL ASSET ACTIVATION FRAMEWORK

Team: Panic! At the Workplace
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Submission Category: Scientific Progress & Stronger Communities

Kalinga AI activates billions in idle government medical equipment to deliver specialist-verified maternal triage to the women most likely to die without it.

1. Target Audience & Beneficiaries

Kalinga AI serves three interconnected groups:

B2C: Rural Pregnant Women (Primary Beneficiaries): Women aged 15–49 living in Geographically Isolated and Disadvantaged Areas (GIDA): remote barangays reachable only by boat, mountain trail, or hours of land travel. They are below the poverty line, have limited education and digital literacy, speak regional dialects, and have no practical access to an OB-Gyne. They are not invisible by choice, but by geography.

B2B: Barangay Health Workers (Operational Partners): The midwives and nurses already stationed in Barangay Health Centers (BHC) and Barangay Health Stations (BHS) are our frontline users. Kalinga AI is built around them: not to replace their judgment, but to multiply their capability. The Philippine Obstetrical and Gynecological Society (POGS) provides the remote specialist verification network.

B2G: Government Health Infrastructure: We operate within the DOH's Health Facilities Enhancement Program (HFEP) and PhilHealth frameworks, deploying through government-operated rural BHCs and BHSs where DOH-procured ultrasound equipment currently sits unused.

2. Problem Statement & Current Gaps

In 2021, **2,478 Filipino mothers died** from pregnancy-related causes, 6–7 women every day [1]. The leading killer is **preeclampsia/eclampsia**: manageable when caught early, often silent until fatal [2].Beyond preeclampsia, the same diagnostic gap leaves conditions like placenta previa, fetal malpresentation, and growth restriction undetected until labor, when options are critically limited.

If unresolved:

- **Human:** Continued preventable deaths.
- **Economic:** A single emergency trip costs rural families ₱5,000–6,000, often exceeding annual savings [3]. Nationally, lost productivity and social support cost hundreds of millions yearly [1].
- **Inequity:** A woman in Makati City walks to an OB-Gyne; a woman in a remote island barangay travels a full day by boat to reach a functioning ultrasound.

The Infrastructure Paradox

DOH has spent billions on portable ultrasound devices for rural facilities, yet **only 200 of 600 HFEP-funded health centers were operational as of late 2025** due to specialist shortages [4], [5]. Idle equipment risks becoming hazardous e-waste without ever saving a life.

Why Current Solutions Fall Short

Approach	Limitation
Manual palpation	Cannot detect placental problems, measure fetal blood flow, or produce a preeclampsia risk score.
Live video telemedicine (e.g., Project MAMA)	Requires high connectivity (Starlink) and extensive midwife training; pilot scale is limited and difficult to scale nationally due to satellite reliance.
General maternal health apps	Primarily urban-focused; provide general health information rather than clinical-grade diagnostic assessment.
High-resource global models (e.g., IN-PoCUS)	Designed for doctor-led systems; lack optimization for remote, midwife-led Philippine health centers.

The Real Gap

The problem is **not** a lack of equipment or coverage. DOH-procured ultrasound devices exist. PhilHealth covers community consultations. Barangay Health Centers have staff. What is missing is the **bridge** that allows a trained midwife, without specialist credentials, to produce a **specialist-grade diagnostic result**. That is the gap **Kalinga AI** is built to close.

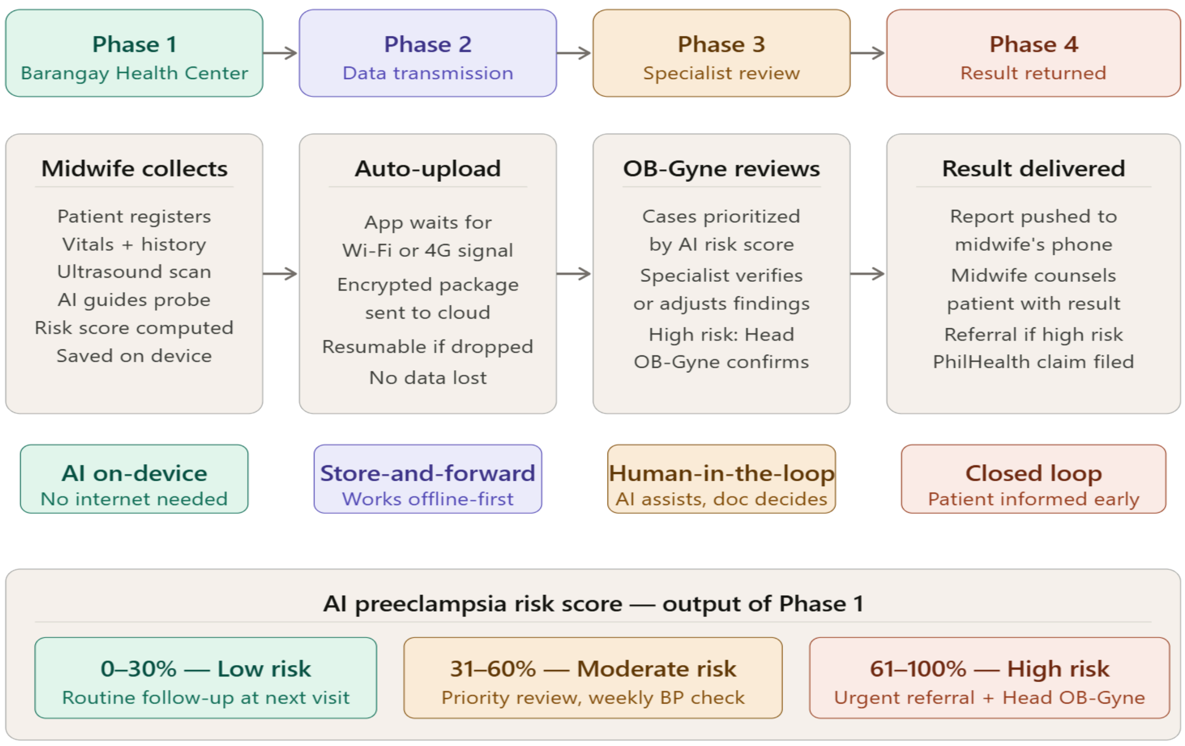
No existing solution combines offline-first edge AI, asynchronous store-and-forward triage, on-device multi-condition maternal risk scoring risk scoring, and government asset activation into a single midwife-operable workflow. Kalinga AI is not an app, it is the missing connective layer between existing infrastructure and specialist-grade outcomes.

3. Kalinga AI: How It Works

Kalinga AI is an **offline-first, AI-assisted diagnostic triage system** that allows a rural midwife (with no specialist training) to capture, process, and transmit a clinical-quality maternal health assessment to a verified OB-Gyne, and receive a specialist-confirmed result; all without needing a continuous internet connection.

Think of it as a "store-and-forward" pipeline: the midwife collects data, the AI processes it locally on the device, the package is sent to the cloud when signal is available, a specialist reviews and confirms it, and the result is returned to the midwife to share with the patient.

Critically, the AI never makes the final call. Every result is verified by a qualified OB-Gyne before it reaches the patient.



Why AI Is the Right Tool Here

- **Pattern recognition at scale:** AI synthesizes dozens of variables (BP, history, ultrasound, gestational age) into a calibrated maternal risk assessment — flagging preeclampsia, placental abnormalities, fetal malpresentation, and growth restriction — in seconds on a smartphone.
- **Automation without replacement:** Elevates midwives to capture diagnostic-quality data; allows one OB-Gyne to review dozens of remote cases per day asynchronously.
- **Early prediction:** Detects risk signals before symptoms appear, enabling intervention while outcomes can still be changed.

Core AI Components

- **Computer Vision (CNN):** On-device, gives real-time audio/visual guidance during sweep (e.g., "angle too steep," "good capture"); ensures clinically useful images from untrained operators.

- **Deep Learning (FetalCLIP):** Identifies anatomical structures, estimates gestational age, flags high-risk markers (e.g., placental position, umbilical flow) [7].
- **Preeclampsia Risk Prediction Model:** Combines BP, BMI, history, risk factors, and ultrasound data into a 0–100% probability score with three-tier risk classification; continuously refined with de-identified, specialist-verified field data.

Data Sources: Trained on FetalCLIP [7] and NatalIA PBF-US1 [8]; governed by NPC Advisory 2024.12.19 [10] and equitable AI frameworks [11].

4. Feasibility & Implementation

Kalinga AI leverages existing assets: DOH-procured ultrasound probes (identified via a Month 0–1 hardware audit of HFEP sites), midwives already stationed in BHCs, and POGS specialists who review cases in under 60 seconds via the Kalinga Dashboard [6]. Triage outputs map directly to PhilHealth's Konsulta package claims [12], and all data handling complies with NPC Advisory 2024.12.19 under the Data Privacy Act [10]. In practice, informed consent is obtained by the attending midwife before registration, using a plain-language verbal explanation delivered in the patient's local dialect, supported by a printed one-page consent summary with visual icons for low-literacy patients. The patient is told explicitly: what data is collected, that images will be reviewed by a remote doctor, and that all records are stored securely and never sold or shared. Consent is logged digitally in the app before the scan can proceed. All data is encrypted, de-identified, and contained within the clinical chain.

Implementation Timeline

Period	Milestone
Month 0–1	Hardware audit of HFEP BHC sites; POGS partnership formalization
Month 1–3	App development; on-device AI deployment; specialist dashboard build
Month 3–4	Pilot rollout in 3–5 rural BHCs; midwife certification training
Month 4–6	System validation; specialist verification workflow testing
Month 6–12	Phased provincial expansion; model improvement via field data
Year 2+	National rollout; ASEAN expansion (Indonesia/Vietnam)

5. Scalability & Impact

Kalinga AI creates a self-sustaining diagnostic network by activating idle ultrasound probes with existing health staff. Its offline-first design enables nationwide scaling with minimal infrastructure investment.

By scaling to 30% of rural health centers, the system can screen up to 200,000 pregnancies annually, preventing hundreds of maternal deaths. This activation of public assets also prevents billions in potential e-waste while providing DOH with real-time rural health data. The architecture is directly replicable in neighboring countries like Indonesia and Vietnam facing similar specialist shortages.

The framework ensures sustainability through PhilHealth Konsulta reimbursements, creating a revenue loop that funds operations. Technical growth is automated as every verified case improves the retraining pipeline, embedding the system into national health infrastructure rather than relying on temporary donor funding.

6. Community Outreach Plan

We build trust through **presence, demonstration, and local endorsers**, not advertising.

- **Track 1 – Workshops (300 health workers):** 3-day certification for midwives/BHWs. Covers: AI basics (plain language), guided scan use, risk-score interpretation, mandatory OB-Gyne verification, and patient communication. Aligned with DOH/POGS standards.
- **Track 2 – “Heartbeat” Demo Days (700 families):** Live demonstrations at BHCs/plazas, co-hosted with barangay captains. A midwife does a live guided scan, shows the AI risk score, and explains doctor verification. Uses the “Magnet Effect” to boost antenatal care compliance [6].
- **Track 3 – Validation:** Framed under DOH’s Safe Motherhood Program and PhilHealth’s Konsulta package. Formal endorsements from Municipal Health Offices, POGS, and DOH Regional Offices. Barangay captains and mayors attend demos as visible endorsers. Success will be measured by: (1) number of midwives certified, (2) number of demo-day attendees recorded via barangay sign-in sheets, and (3) a pre/post AI awareness survey administered to all workshop participants.

Period	Activity	Target Reach
Month 0–1	Partnership signing with POGS and MHOs; site selection	Stakeholders
Month 1–2	First cohort of midwife certification workshops	60 Health Workers
Month 2–3	First "Heartbeat" Demo Days in pilot barangays	200 Families
Month 3–6	Expanded certification + demo days across 5 municipalities	500+ Members
Month 6–12	Full provincial rollout; health assemblies; LGU briefings	1,000+ Reached

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